

Summary of Solar Radiation Modification (SRM) Impacts on Southeast Asia and Eastern Africa

Abstract

This document provides a summary of two research papers analyzing the impacts of Solar Radiation Modification (SRM) interventions on extreme climate events. The first paper focuses on temperature extremes and heatwaves in Southeast Asia, while the second evaluates temperature and rainfall extremes in Eastern Africa. Special attention is given to the mathematical expressions, algorithms, and indices used in their methodologies.

1 Impacts of SRM on Temperature Extremes and Heatwaves in Southeast Asia

This study evaluates the effects of SRM (specifically G6Solar and G6Sulfur scenarios) alongside traditional emission pathways (SSP245 and SSP585) using downscaled and bias-corrected GeoMIP6 datasets.

1.1 Mathematical Formulations and Algorithms

1.1.1 Bilinear Interpolation

To address discrepancies in grid resolutions between simulation outputs and observed datasets (ERA5), bilinear interpolation was utilized. The interpolated temperature $T'_{(x,y)}$ at a new point (x, y) is calculated based on the four nearest grid points:

$$T'_{(x,y)} = \frac{1}{(x_2 - x_1)(y_2 - y_1)} \left[(x_2 - x)(y_2 - y)T_{(x_1,y_1)} + (x - x_1)(y_2 - y)T_{(x_2,y_1)} + (x_2 - x)(y - y_1)T_{(x_1,y_2)} + (x - x_1)(y - y_1)T_{(x_2,y_2)} \right] \quad (1)$$

where $T_{(x_1,y_1)}$, $T_{(x_2,y_1)}$, $T_{(x_1,y_2)}$, and $T_{(x_2,y_2)}$ represent the temperature values at the surrounding grid points, bounding the cell containing (x, y) .

1.1.2 Bias Correction (Delta Method)

A linear scaling approach based on the delta method was applied to align the interpolated simulation data with observations. For each grid point i and month m over the climatology period, a bias adjustment factor $A_{(i,m)}$ is defined as:

$$A_{(i,m)} = \overline{\text{Observed}}_{(i,m)} - \overline{\text{Interpolated Simulation}}_{(i,m)} \quad (2)$$

The bias-corrected temperature $T'_{(i,m)}$ is then computed by applying this adjustment factor:

$$T'_{(i,m)} = T_{(i,m)} + A_{(i,m)} \quad (3)$$

1.1.3 Algorithm: Downscaling and Bias Correction

The procedures for downscaling and bias correction are summarized in Algorithm 1.

Algorithm 1 Downscaling and Bias Correction Process

Require: Simulation data S , Observed data (ERA5) O , Climatology period P_{clim}

Ensure: Bias-corrected simulation dataset $S_{corrected}$

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1: Step 1: Bilinear Interpolation Downscaling
2: for each grid point  $(x, y)$  in  $S$  do
3:   Identify the four nearest bounding grid points:  $(x_1, y_1), (x_2, y_1), (x_1, y_2), (x_2, y_2)$ 
4:   Calculate interpolated temperature  $T'_{(x,y)}$  using Equation (1)
5: end for
6: Let  $S_{interp}$  denote the interpolated dataset
7: Step 2: Delta Method Bias Correction
8: for each grid point  $i$  and month  $m$  in  $P_{clim}$  do
9:   Compute observed mean:  $\overline{O}_{(i,m)}$ 
10:  Compute simulated mean:  $\overline{S}_{interp(i,m)}$ 
11:  Compute bias adjustment factor:  $A_{(i,m)} = \overline{O}_{(i,m)} - \overline{S}_{interp(i,m)}$ 
12: end for
13: for each grid point  $i$  and month  $m$  in future simulation scenarios do
14:   Apply bias correction:  $T'_{(i,m)} \leftarrow T_{(i,m)} + A_{(i,m)}$ 
15: end for
16: return  $S_{corrected}$ 

```

1.1.4 Temperature and Heatwave Indices

A relative, percentile-based threshold (90th percentile of T_{max} during the baseline period) was used to define heatwaves. The study evaluated several indices:

- **TXx:** Annual maximum daily maximum temperature (°C).
- **TNx:** Annual maximum daily minimum temperature (°C).
- **HWN (Heatwave Number):** Total number of heatwave events within a year.
- **HWF (Heatwave Frequency):** Frequency of heatwave days per year.
- **HWD (Heatwave Duration):** Longest duration of heatwave events per year.
- **HWA (Heatwave Amplitude):** Peak daily value in the hottest heatwave events per year (°C).

2 Simulated Response of the Climate of Eastern Africa to Stratospheric Aerosol Intervention

This study assesses the potential of SRM via Stratospheric Aerosol Injection (SAI) using the CESM2-WACCM6 model under the ARISE-SAI project, comparing it against the SSP2-4.5 emissions pathway.

2.1 Mathematical Formulations and Algorithms

2.1.1 Temperature and Rainfall Extremes Definition

The analysis focuses on several climate extreme indices to capture extreme temperature and precipitation events:

- **WSDI (Warm Spell Duration Index):** Measures consecutive days when the daily maximum temperature exceeds the local and seasonal 90th percentile (calculated using a 5-day moving window).
- **DTR (Daily Temperature Range):** The mean difference between daily maximum and minimum temperatures.
- **CDD (Consecutive Dry Days):** Maximum number of consecutive days with less than 1 mm of precipitation.
- **CWD (Consecutive Wet Days):** Maximum number of consecutive days with more than 1 mm of precipitation.

2.1.2 Standardized Precipitation-Evapotranspiration Index (SPEI)

The SPEI is utilized to assess the frequency of extreme wet and dry events. It improves upon the Standardized Precipitation Index (SPI) by incorporating potential evapotranspiration (PET), calculated using the simplified Thornthwaite approach. The SPEI is defined as the difference between precipitation and PET. Table of extreme event categories based on SPEI thresholds:

SPEI Threshold	Extreme Event Category
$1.83 \leq \text{SPEI}$	Extremely wet
$1.42 < \text{SPEI} \leq 1.82$	Severely wet
$1.0 < \text{SPEI} \leq 1.42$	Moderately wet
$-1.0 \leq \text{SPEI} \leq 1.0$	Near Normal
$-1.42 < \text{SPEI} \leq -1.0$	Moderate drought
$-1.82 < \text{SPEI} \leq -1.42$	Severe drought
$\text{SPEI} \leq -1.82$	Extreme drought

2.1.3 Algorithm: SPEI Calculation

The computational flow for classifying extreme events based on SPEI is outlined in Algorithm 2.

Algorithm 2 Calculation and Classification of SPEI

Require: Monthly Precipitation P , Monthly Temperature T

Ensure: SPEI Classification for Extreme Events

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1: for each time step  $t$  do
2:   Compute Potential Evapotranspiration  $PET_t$  using simplified Thornthwaite ap-
   proach from  $T_t$ 
3:   Calculate climatic water balance:  $D_t = P_t - PET_t$ 
4: end for
5: Accumulate  $D_t$  over different time scales (e.g., 1, 3, 6, 12 months)
6: Fit the accumulated values to a probability distribution function
7: Determine the cumulative probability  $F(D)$ 
8: Transform  $F(D)$  into a standard normal distribution to yield the SPEI value
9: if  $SPEI \geq 1.83$  then
10:   Classify as Extremely Wet
11: else if  $1.42 < SPEI \leq 1.82$  then
12:   Classify as Severely Wet
13: else if  $-1.42 < SPEI \leq -1.0$  then
14:   Classify as Moderate Drought
15:   ... (continue based on SPEI Thresholds Table)
16: end if
17: return SPEI Classification
```

3 Conclusion

Both papers demonstrate that SRM interventions (G6Solar, G6Sulfur, and ARISE-SAI) have a significant moderating effect on extreme temperature increases compared to high-emission scenarios (SSP585 and SSP2-4.5). However, SRM impacts on precipitation and specific heatwave characteristics exhibit significant spatial and seasonal variability. In Southeast Asia, SRM interventions effectively reduce heatwave duration, amplitude, and frequency. In Eastern Africa, SAI helps stabilize surface temperatures but introduces varied impacts on consecutive dry/wet days and extreme rainfall frequencies depending on the season and specific region.